dynamic combat scenarios without direct human intervention. Despite major advancements in AI, limitations persist in contextual reasoning, real-time decision-making, and data interpretation under uncertain environments. Furthermore, the integration of multi-sensor fusion, high-speed data processing, and secure communications within a compact UAV framework poses difficulties in terms of hardware optimization, thermal management, and power efficiency.

Stealth and low-observable (LO) designs also introduce manufacturing challenges, as they require precise fabrication of composite materials and radar-absorbent coatings. These processes are costly and time-consuming, and the degradation of stealth coatings over time can compromise mission effectiveness.

5.7.2 Operational limitations

Fourth-generation UAVs depend heavily on data networks and satellite links for coordinated operations, particularly in manned–unmanned teaming (MUM-T) and swarm systems. Any disruption in communication due to jamming, cyberattacks, or environmental interference can cause mission failure or loss of control. Additionally, the complexity of maintenance for AI-integrated drones, coupled with their limited field reparability, increases logistical burdens.

Operational endurance is another limitation. Despite advancements in hybrid propulsion and energy management, the balance between payload weight, fuel efficiency, and range remains constrained. Harsh environments such as deserts, oceans, or high altitudes further test the reliability of propulsion systems, sensors, and onboard electronics.